



EduTutor AI: Personalized Learning

An AI-powered learning assistant that transforms digital education through intelligent tutoring and adaptive learning paths.

Team Members:

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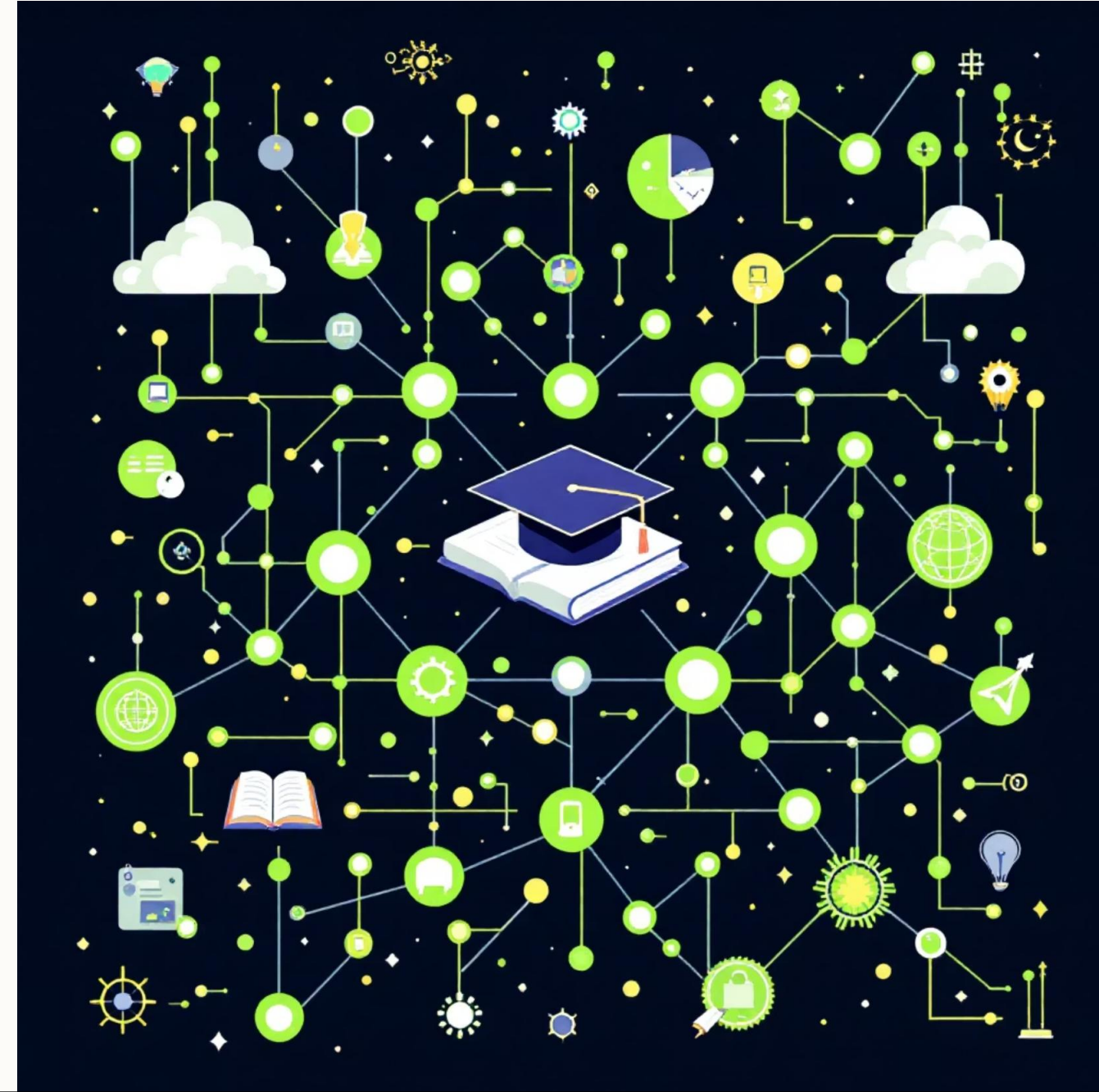
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Project Overview

- ❖ Rapid growth of AI in education technology (EdTech)
- ❖ Need for tools that explain, quiz, and summarize dynamically
- ❖ Edu Tutor AI bridges this gap through automation
- ❖ Provides personalized, on-demand educational assistance
- ❖ Enhances student engagement and independent learning

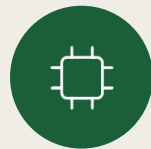


Model Setup



NLP

Ensures high-quality natural language generation



IBM Granite Models

`ibm-granite/granite-3.2-2b-instruct`



Python Programming

Proficiency in Python for development and implementation



Version Control

Git knowledge for project management and collaboration



Google Colab GPU

Understanding of T4 GPU usage in Colab environment

Educational Features



Concept Explanation

Detailed breakdown with examples



Summarization

Concise bullet point summaries



Flashcards

Q&A-style flashcards

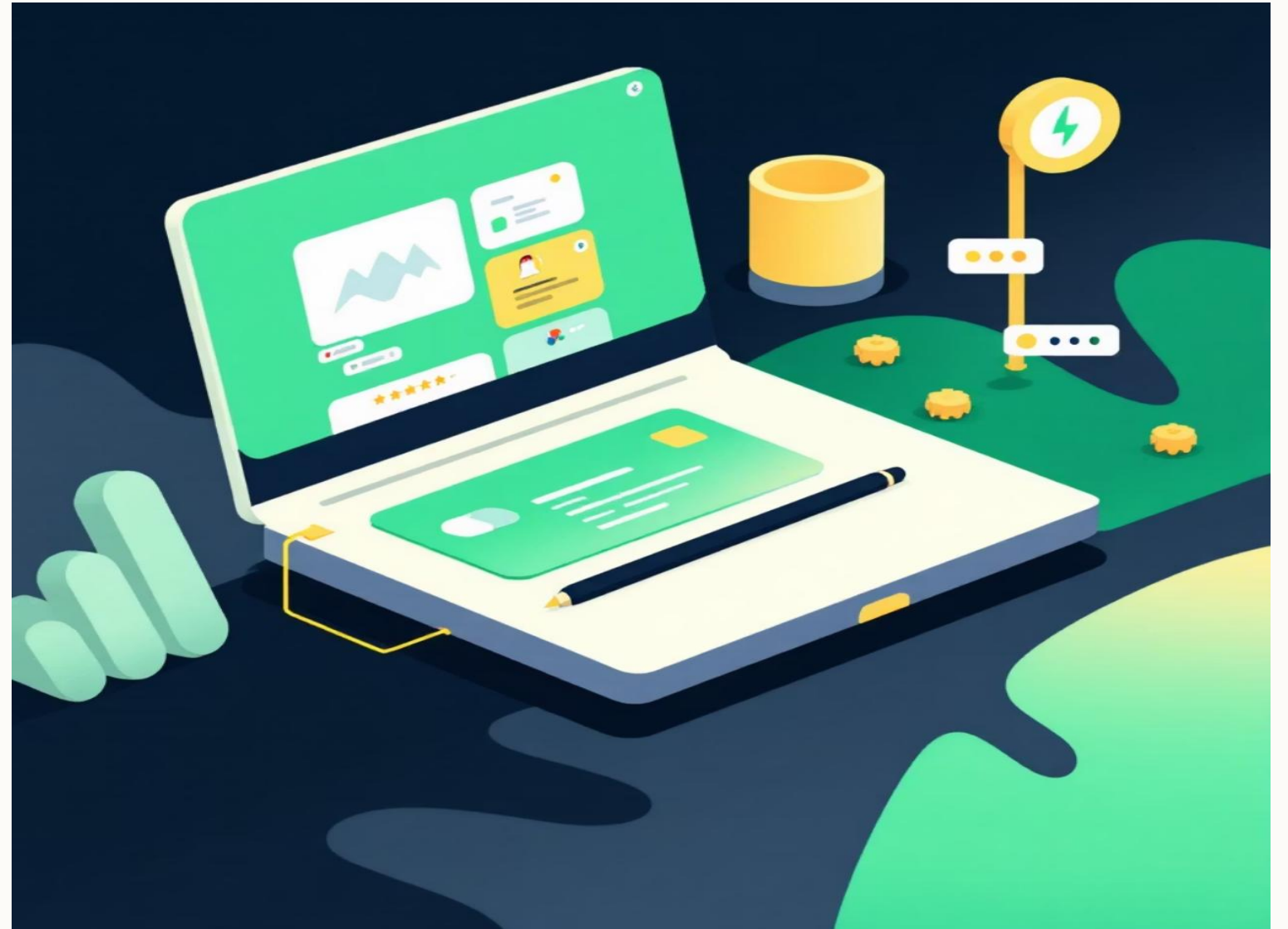


Quiz Generator

varied questions (MCQ, T/F, short answers)

VOICE AND AUDIO INTERACTION

- Converts **speech to text** for concept entry
- Generates responses and explains through text
- Uses **gTTS** to convert text back to **audio**
- Enables multi-format learning (reading +listening)



Application Output: Educational AI Assistant

The application provides an AI-powered educational interface with modular tabs for concept explanation, summaries, and learning tools.

 Educational AI Assistant

Concept Explanation

Summary

Flashcards

Quiz Generator

Enter a concept

Machine Learning

Explain

Explanation

Machine Learning (ML) is a subfield of artificial intelligence (AI) that involves training models on data to make predictions or decisions without being explicitly programmed for every single scenario. It allows computers to learn automatically from patterns and statistical insights derived from the data, improving their performance over time as they encounter more examples.

There are three primary types of machine learning: supervised learning, unsupervised learning, and reinforcement learning. Each approach has distinct characteristics and applications.

1. Supervised Learning: This method relies on labeled data, meaning the input features are accompanied by corresponding output values or 'labels'. The goal is to learn a mapping function from inputs to outputs. For example, consider a simple email spam classifier:

- Input features (unlabeled data): Words present in an email (e.g., "free," "prize," "urgent")
- Output labels: Whether the email is spam (labeled data) or not (ham)

Concept Explanation

Enter a topic and receive AI-generated, easy-to-understand educational explanations.

In summary, machine learning is a powerful tool enabling computers to learn and improve from experience. By understanding different machine learning techniques and their applications, we can unlock innovative solutions to a wide range of real-world problems.

Listen

Download Explanation

Audio



0:005:22



File

output.txt3.8 KB

Audio Explanation

Listen or download the generated explanation as audio or text for flexible learning.

Downloaded File

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- Input features (unlabeled data): Words present in an email (e.g., "free," "prize," "urgent")
- Output labels: Whether the email is spam (labeled data) or not (ham)
- Training: The algorithm learns to associate certain words with the spam label by processing many examples of emails.
- Prediction: Once trained, the model can classify new, unseen emails as spam or ham based on the learned pattern.

2. Unsupervised Learning: In this technique, the model is given only the input features and no explicit output labels. Instead, it seeks to discover inherent structures or patterns within the data by itself. An example is clustering customers based on their purchasing behavior:

- Input features: Customer transaction history (e.g., products bought, frequency, time)
- No explicit output labels
- Training: The algorithm identifies similar groups or clusters of customers who share common purchasing tendencies
- Application: Retail companies can tailor marketing efforts, product recommendations, and personalized offers to these distinct customer segments

3. Reinforcement Learning: Unlike supervised and unsupervised learning, reinforcement learning operates in an interactive environment where an agent learns to maximize a cumulative reward signal through trial and error. A classic example is training a computer program to play a game like chess or Go:

- State: The current game board configuration
- Action: Moves the program makes in the game
- Reward: The program receives points for winning the game and loses points for losing or drawing
- Training: The AI agent refines its strategy by repeatedly playing games against itself or human opponents, learning from both victories and defeats
- Application: Reinforcement learning has led to significant advancements in game-playing AI, enabling systems to learn complex strategies and master new games

Besides these primary types, there are also ensemble methods (like Random Forests and Gradient Boosting) and deep learning techniques, which combine various ML algorithms and neural networks to further enhance predictive capabilities.

Enter a topic to get an in-depth explanation of core concepts.

Summary Generator

Educational AI Assistant

-  Concept Explanation
-  **Summary**
-  Flashcards
-  Quiz Generator

Enter a topic

Neural Network

Summarize

Summary

1. **Neural Network** is a computing system inspired by the human brain's biological neural networks. It's designed to recognize patterns, learn from data, and make decisions based on that learning.

2. **Architecture**: Consists of multiple layers of interconnected nodes or 'neurons'. Input layer receives data, hidden layers process information, and output layer delivers the final result.

3. **Roles**:

 - **Learning**: Adjusts internal parameters (weights and biases) to minimize error during training.
 - **Generalization**: Applies learned knowledge to unseen data, enabling prediction and decision-making.
 - **Optimization**: Uses algorithms like backpropagation to optimize the network's performance.

4. **Types**:

Instant summary generation for quick understanding of a topic.

Flashcards Generator

 Educational AI Assistant

 Concept Explanation

 Summary

 **Flashcards**

 Quiz Generator

Enter a topic

Data Science

Generate Flashcards

Flashcards

Generate 5 flashcards for Data Science in 'Question:... | Answer:...' format.

1.

Question: What is the primary goal of Data Science?

Answer: The primary goal of Data Science is to extract meaningful insights and knowledge from structured and unstructured data through a systematic combination of methods, processes, and algorithms.

2.

Question: What is the fundamental component of a Data Science project?

Answer: The fundamental component of a Data Science project is data, which can be in various forms like structured (databases), semi-structured (JSON, XML), or unstructured (text, images, videos).

Generate topic-wise educational flashcards for studying.

Quiz Generator

 Educational AI Assistant

 Concept Explanation

 Summary

 Flashcards

 **Quiz Generator**

Enter a topic

Artificial intelligence

Generate Quiz

Quiz Questions

1. Multiple Choice (MCQ): Which of the following is NOT a primary function of Artificial Intelligence?

- a) Data processing
- b) Learning from experience
- c) Emotional intelligence
- d) Creating physical objects

2. True/False: Machine learning, a subset of AI, enables systems to improve their performance on a specific task by learning from the data, without being explicitly programmed.

3. Short Answer: Explain the concept of "Deep Learning" in the context of AI. Deep learning refers to a subset of machine learning techniques that are inspired by the structure and function of the human brain, specifically neural networks with many layers (hence 'deep'). These networks can learn and represent complex patterns and hierarchies from raw data, such as images, audio, or text.

Automatically generate quizzes based on any AI/tech topic.



EduTutor AI Overview

With our environment configured, begin coding our AI tutor. The implementation includes installing required libraries, loading the IBM Granite model, and creating an interactive Gradio interface.

1

Install Dependencies

Import Gradio, transformers, and required libraries

2

Load Model

Initialize the granite-3.2-2b-instruct model from Hugging Face

3

Create Interface

Build Gradio UI for student-tutor interactions

4

Test & Deploy

Run the application and verify functionality

CONCLUSION & FUTURE SCOPE

Conclusion:

- EduTutor simplifies learning by providing personalized study resources.
- Efficiently tracks student progress and identifies weak areas.
- Enhances learning engagement through interactive content.

Future Scope:

- Integrate AI-driven doubt solving and recommendation systems.
- Add multi-language support for wider accessibility.
- Include gamification features to motivate students.
- Expand to support collaborative learning and peer interactions.

